Leakage-Free Benchmarking of ALL vs. AML Leukemia Subtype Classification Across Three Gene-Expression Cohorts

Interim Report: Literature Review, Data Sets and Exploratory Data Analysis

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***Abstract*—Acute lymphoblastic (ALL) and acute myeloid leukemia (AML) need different therapies, and the Golub et al. microarray data set (72 patients, 7,129 probes) is the classic benchmark for separating them from gene expression. Because the data are tiny and very high-dimensional ($p \gg n$), many published accuracies near 100% rest on validation designs that leak information. This interim report lays the foundation for a leakage-free comparison of machine learning (ML) methods. We (i) review 38 verified references with a documented search protocol, (ii) select two independent open cohorts with the same clinical question—the multicentre MILE study (GSE13159, 1,292 ALL/AML samples, Affymetrix HG-U133 Plus 2.0) and a paediatric cohort of Andersson et al. (GSE7186, 121 samples, two-colour cDNA arrays)—and (iii) run a fully scripted exploratory analysis. The Golub data contain no missing values but 28.7% negative intensities, which motivates clipping and a $\log_2$ transform; after that, PC1 alone separates the classes with an AUC of 0.984. A shift between the training and independent sets is measurable. Gene-wise t statistics correlate across cohorts (Spearman $\rho = 0.56$ Golub–MILE, $\rho = 0.44$ Golub–Andersson), and dataset identity, not class, dominates the pooled variance until each cohort is standardised. These findings fix the design of the final study: in-fold feature selection, nested cross-validation and per-cohort normalisation.**

***Index Terms*—leukemia, gene expression, microarray, exploratory data analysis, batch effect, nested cross-validation, small-sample learning**

I. Introduction

Clinical problem. ALL arises from lymphoid and AML from myeloid progenitor cells. Both present with similar symptoms and blasts in bone marrow, yet induction chemotherapy, risk stratification and transplant decisions differ substantially, so an incorrect lineage assignment directly affects treatment. In routine practice the distinction relies on morphology, cytochemistry, immunophenotyping and cytogenetics; gene-expression profiling was proposed as a single, objective assay that reproduces these labels and can reveal sub-classes [1], [2]. Throughout this work the task is called *ALL vs. AML subtype classification*: both classes are leukemias, so the label "non-cancer" used in the course module for ALL is biologically incorrect and is not used here.

Video: https://youtu.be/VIDEO-LINK-HERE. Code: https://github.com/mrayenesay/leukemia-all-aml-ml. Every number in this report is generated by `scripts/run_eda.py` and imported as a LaTeX macro; see Section VI.

Microarray technology. A DNA microarray measures the abundance of thousands of mRNA transcripts in parallel by hybridising labelled cDNA/cRNA to immobilised probes. Affymetrix oligonucleotide arrays (used by Golub and MILE) report one intensity per probe set; the older MAS 4 "average difference" (perfect-match minus mismatch) can be negative and is accompanied by a Present/Marginal/Absent (P/M/A) detection call. Two-colour spotted cDNA arrays (Andersson) co-hybridise the sample (Cy3) with a common reference RNA (Cy5) and report a log ratio.

Research questions. The project tests four falsifiable questions:

- **RQ1** (method comparison): Under a leakage-free 5×5 nested stratified cross-validation (CV) on all 72 Golub samples, do penalised logistic regression, a support vector machine (SVM) and a random forest differ in balanced accuracy? H_0: equal mean balanced accuracy (Friedman test, corrected resampled t-test [3], [4]).
- **RQ2** (protocol): Does the single 38/34 split of the course module yield an estimate inside the 95% confidence interval of nested CV, or is it optimistic or pessimistic?
- **RQ3** (generalisability): Is the ALL/AML signal shared across cohorts and platforms, i.e. is the correlation of gene-wise test statistics $\rho > 0$ and is the top-100 gene overlap larger than chance? In the final report: does a model trained on one cohort keep balanced accuracy ≥ 0.90 on another?
- **RQ4** (bonus): Do a deep model (MLP/autoencoder) and a tabular transformer (TabPFN) outperform the best classical model under the identical protocol?

RQ3 is answered in part already by the exploratory analysis in Section IV.

Contributions of this interim report. (1) a documented review of 38 verified references (Sec. II); (2) two harmonised external cohorts, three data sets in total (Sec. III); (3) a scripted

exploratory data analysis (EDA) with numbered modelling implications (Sec. IV); (4) a pre-registered plan for the final report (Sec. V).

II. Literature Review

Search protocol. We queried Crossref (keyword search filtered to journal articles), Europe PMC (full text) and PubMed/NCBI E-utilities on 3 Nov 2026. Discovery was restricted to 2021–2026; classic papers (1995–2020) were added by identifier and verified the same way. Queries were, e.g., *"golub leukemia gene expression classification"*, *"ALL AML leukemia microarray feature selection classification"*, *"machine learning cancer classification gene expression data review"* and the full-text query *"7129" AND "72 samples" AND (leukemia OR leukaemia) AND classification*. *Inclusion:* peer-reviewed journal or conference papers about the data sets used, classification or feature selection on expression data, validation bias, batch effects, or recent small-sample tabular learning. For papers with a reported Golub result, the full text had to confirm the 72×7129 data. *Exclusion:* conference abstracts, image-based leukemia studies, papers whose leukemia data were not the Golub set, and unindexed venues. Of 152 records screened by title, 24 abstracts and 5 full texts were assessed; 30 studies were included, plus 8 method/statistics references cited in the analysis plan. All 38 entries are peer-reviewed and 11 were published in 2021 or later. Every DOI was checked against the Crossref record (title, authors, venue, year); the log is `reports/literature/search_log.md`.

Data sets and early studies. Golub et al. [1] introduced class discovery and class prediction with a 50-gene weighted-voting predictor learnt on 38 samples and applied to 34 independent samples. Armstrong et al. [5] showed that MLL-rearranged leukemias form a distinct expression class, and the MILE study [2], [6] standardised expression profiling across laboratories, reaching 92.2% accuracy over 18 classes. Andersson et al. [7] classified 121 consecutive childhood leukemias with k-nearest neighbours. CuMiDa [8] curates such data sets for benchmarking.

Classifiers and gene selection. SVMs [9] became the default for microarrays [10], and SVM-RFE introduced wrapper gene selection [11]. Dudoit et al. [12] found that simple diagonal discriminant analysis and k-NN rival complex methods, Tibshirani et al. [13] proposed nearest shrunken centroids, and Díaz-Uriarte and Alvarez de Andrés [14] used random forests [15] for gene selection (.632+ error on Golub: SVM 0.014, RF 0.051). Statnikov et al. [16] found multicategory SVMs strongest across 11 data sets, and the elastic net [17] offers sparse, correlated-feature-aware linear models. Reviews from 2022–2023 summarise hybrid filter–wrapper selection as the dominant approach [18], [19].

Validation bias and batch effects. Selecting genes on all samples before CV produces optimistic error rates [20], and so does tuning hyper-parameters on the evaluation folds; nested CV removes this bias [21], [22]. Inference on CV estimates needs a variance correction for overlapping training sets [3], and multiple classifiers should be compared with rank tests [4]. Batch effects can dominate expression data [23] and are commonly corrected with ComBat [24].

Recent work. Large-scale transcriptome studies show that AML can be predicted across 105 studies [25] and that decentralised "swarm" learning works on multi-site leukemia data [26]. Recent Golub results are very high [27]–[30], but in our reading two of them rank genes outside the CV loop or rely on random hold-out splits only (Table I). For tabular data, tree ensembles often beat deep networks [31]; FT-Transformer [32] and the tabular foundation model TabPFN [33] are the strongest neural alternatives, and GenePT [34] builds gene embeddings from language-model text representations.

Gap. Reported Golub accuracies range from 97% to almost 100%, but they are hard to compare: protocols mix LOOCV, k-fold CV, the fixed 38/34 split and random hold-outs. Gene selection is often done on the full data, and confidence intervals are rarely reported. Moreover, almost all studies evaluate on Golub alone, so it remains unclear whether the selected genes or the method rankings transfer to other platforms and patient groups. This project addresses all three issues: one leakage-free nested protocol for every method, interval estimates with statistical tests, and the same pipeline applied to two external cohorts.

III. Data Sets

Table II summarises the three data sets; their class distributions are plotted in Appendix C (Fig. 7).

Golub (main data set). Golub et al. [1] (DOI 10.1126/science.286.5439.531), obtained from the Kaggle mirror of the original Broad Institute files ("Gene expression data set (Golub et al.)", licence CC0). Platform: Affymetrix Hu6800/HuGeneFL (GEO GPL80), 7,129 probe sets (58 AFFX controls; 6,434 map to a gene symbol), MAS 4 average difference with P/M/A calls. Mostly adult patients and bone-marrow samples; the independent set is more heterogeneous (it also contains peripheral-blood samples) [1]. The reference split has 38 training samples (27 ALL / 11 AML) and 34 independent samples (20 / 14), 65%/35% overall.

MILE (similar data set 1). Microarray Innovations in LEukemia, stage 1, GEO GSE13159 (DOIs 10.1200/JCO.2009.23.4732, 10.1111/j.1365-2141.2008.07261.x) [2], [6]. It contains 2,096 diagnostic samples in 18 classes, profiled with standardised protocols in several centres on Affymetrix HG-U133 Plus 2.0 (GPL570, 54,675 probe sets) from quantile-normalised signal [35]. We map the 14 ALL and AML classes to the binary label and exclude CLL, CML, MDS and non-leukemic samples. This leaves 1,292 samples (750 ALL / 542 AML; 1,211 bone marrow, 81 peripheral blood). *Why it was chosen:* same clinical question, a much larger multicentre cohort, Affymetrix technology, and a data set large enough for pre-training (bonus B1).

Andersson (similar data set 2, bonus B4). 121 consecutive childhood acute leukemias from Lund, Sweden, GEO GSE7186 (DOI 10.1038/sj.leu.2404688) [7]: 87 B-lineage ALL, 11 T-cell ALL and 23 AML (6 normal bone-marrow

TABLE I
LITERATURE SUMMARY (SUBSET; THE FULL 30-STUDY TABLE IS IN APPENDIX A). "N/R": NO GOLUB-SPECIFIC NUMBER IN THE TEXT WE COULD ACCESS.

Ref. (year)	Data set	Pre-processing / feature selection	Method	Validation	Best result	Leakage risk
[1] (1999)	Golub 38/34	class-correlation (SNR) gene selection	50-gene weighted voting	fixed independent test	n/r	low (selection on train)
[12] (2002)	Golub, NCI60, lymphoma	clip, log, univariate filter	DLDA, k-NN, trees	repeated 2:1 splits	n/r	low
[20] (2002)	colon, leukemia	selection inside vs. outside CV	SVM-RFE	external 10-fold, .632+	shows bias	– (studies leakage)
[14] (2006)	Golub + 8	RF backward elimination	RF vs. SVM, k-NN	.632+ bootstrap	error: SVM .014, RF .051	low
[2] (2010)	MILE stage I+II	standardised U133+2	multiclass classifier	multi-centre + prospective	92.2% (18 classes)	low
[27] (2022)	Golub + 3	GNN-based ranking	SVM	10-fold CV	acc. 0.971 (2 genes)	unclear
[28] (2022)	Golub + 2	ensemble wrapper (k-NN)	stacked DNN	k-fold CV	acc. 99.6%	unclear (selection order)
[29] (2022)	Golub (UCI)	PSO (2,323 genes)	optimised CNN	random 70/30, 80/20	acc. 99.9%	high (no CV)
[30] (2024)	Golub + CuMiDa	RF importance ≥ 0.04	super learner	10-fold CV + 20% test	acc. 0.98	high (ranking before CV)

TABLE II
THE THREE DATA SETS SIDE BY SIDE.

	Golub	MILE	Andersson
Source	[1], Kaggle (CC0)	GSE13159	GSE7186
Platform	Hu6800 (GPL80)	U133 Plus 2.0 (GPL570)	27K cDNA (GPL4861)
Technology	1-colour oligo	1-colour oligo	2-colour cDNA
Measure	MAS4 avg. diff.	range-scaled log	$\log_2$ ratio
Population	mostly adult	multicentre	paediatric
Samples n	72	1,292	121
Features p	7,129	54,675	21,370
ALL / AML	47 / 25	750 / 542	98 / 23
Split	38 train / 34 test	none	none
Missing values	0	0.08%	7.19%
Genes (symbols)	5,268	20,848	9,677

controls excluded). Most samples are bone marrow (108), the rest peripheral blood (13). Platform: Swegene Human 27K RAP two-colour spotted cDNA array (GPL4861, 21,370 reporters). Values are lowess-normalised $\log_2$(sample/Universal Human Reference) ratios, centred per hospital and pre-filtered for 40% presence, so 7.19% of the values are missing. *Why it was chosen:* same question but a different technology, an all-paediatric population and real missing values, which makes it a demanding external test.

Licences. GEO data are public; NCBI places no restrictions on their use, although submitters may hold rights to the original data. All three sets are used for non-commercial academic research with citation of the original publications.

Harmonisation. Probes were mapped to HGNC symbols with the GEO platform annotations (GPL80/GPL570/GPL4861). Ambiguous probes ("`A///B`") were removed and multiple probes per gene were collapsed to the probe with the highest mean expression. This gives 5,268, 20,848 and 9,677 genes, with 5,207 genes shared by Golub and MILE, 3,487 by Golub and Andersson, and 3,469 by all three. The platforms differ in scale (linear intensity vs. log ratio), probe design and number of genes, so cross-cohort modelling will use per-cohort standardisation on the common genes (Section IV-F).

IV. EXPLORATORY DATA ANALYSIS

All analyses were run in Python (pandas, SciPy, scikit-learn [36], seaborn) by `scripts/run_eda.py` with seed 42. *Note on leakage:* the supervised gene rankings in this section describe the data only. They must never be reused to select features for the models, because that would reproduce the selection bias of [20].

A. Data quality

Golub. The 513,288 values contain 0 missing or infinite entries, so the "fill INF with the mean" step of the module has no effect on the original files (it matters only after a log of non-positive values, which clipping avoids). However, 28.7% of the values are negative and 32.7% lie below 20, while 0.83% exceed 16,000 (range -28,400 to 71,369; Fig. 1a). Negative average differences arise when the mismatch signal exceeds the perfect-match signal, i.e. for transcripts that are absent. The calls confirm this: 70.2% of the measurements are Absent, 28.2% Present and 1.6% Marginal. The median value is 33 for Absent and 662 for Present calls (Fig. 1c), and 1,971 probes are never called Present. The Present rate per array varies from 19.1% to 40.5%, with no visible class pattern (Fig. 1b). **Label mapping:** the patient columns are *not* in numeric order (training order 1–27, 34–38, 28–33). Labels were therefore joined on the patient number, and the script asserts that training IDs are exactly 1–38, test IDs 39–72 and that the class counts equal 27/11 and 20/14.

MILE and Andersson. Although GEO describes the MILE values as normalised signal, every array spans exactly 0.00–1.00. Comparing one array with its ArrayExpress sample table shows that the values are a monotone (Spearman 1.000) and almost linear function of $\log_2$ signal (Pearson 0.998 vs. 0.51 with the linear signal): they are per-array range-scaled log intensities, so no further log transform is applied. 45 probe sets are empty in every array and were dropped (0.08% of all values); there are no infinite entries. Andersson has 7.19% missing values: 12,084 reporters have at least one gap and 2,841 more than 20%; one array lacks 24.3% of its values (Appendix, Fig. 15). Imputation is therefore required, and it must be fitted inside the training folds.

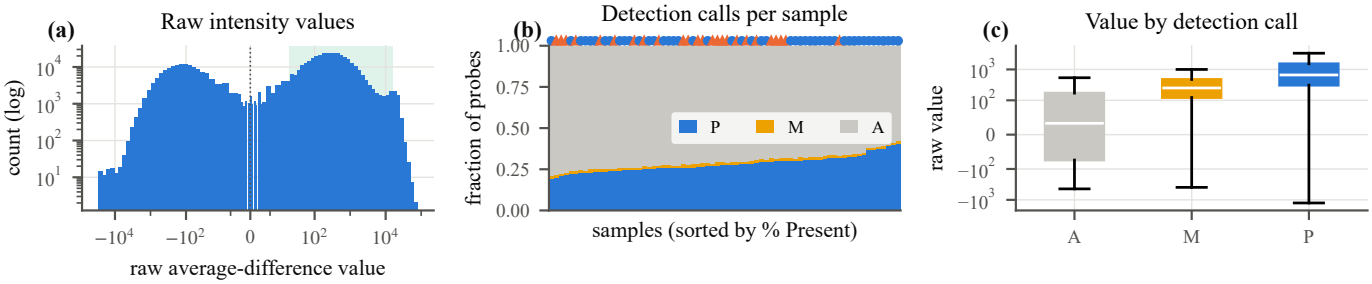


Fig. 1. Golub data quality. (a) Raw average-difference values on a symmetric-log axis; the shaded band is the clipping window [20, 16000]. (b) Fraction of Present/Marginal/Absent calls per array (markers on top: class). (c) Raw value by detection call: Absent calls coincide with negative or near-zero values.

B. Distributions and transformation

Following [12], values were clipped to [20, 16000] and $\log_2$-transformed. The pooled skewness falls from 7.08 to 0.55, and the Spearman correlation between gene mean and standard deviation drops from 0.95 to 0.18 (Appendix, Fig. 8). The transform thus removes most of the mean–variance dependence that would otherwise let a few high-intensity genes dominate distance-based methods (SVM, k-NN, PCA). After clipping, about one third of the values sit at the floor ($\log_2 20$), visible as a spike in the density. A fold-change filter (max/min > 5, max$-$min > 500) keeps 3,859 of the 7,129 probes. MILE (range-scaled log) and Andersson ($\log_2$ ratio, median -0.04) are already on log scales.

C. Dimensionality reduction

PCA on the centred, filtered $\log_2$ data (Fig. 2) shows that the variance is spread out: PC1 explains 10.8% and the first ten PCs 49.6%, and 37 PCs are needed for 80%. Even so, PC1 alone orders ALL and AML with an AUC of 0.984, i.e. the class difference is the largest single source of variation. t-SNE (perplexity 10) [37] shows two clear groups (silhouette 0.40 vs. 0.17 in 10-D PCA space), and the independent samples (black edges) mix with the training samples. In MILE (top 5,000 variable probes) the best single PC reaches AUC 0.987. In Andersson PC1 explains 17.5% with AUC 0.999.

D. Discriminative genes

For each gene we computed Welch's t-test, the Mann–Whitney U test (both with Benjamini–Hochberg FDR [38]) and Golub's signal-to-noise ratio (SNR). In Golub, 945 of 3,859 genes are significant at $q < 0.05$ (567 at $q < 0.01$; 1,014 with Mann–Whitney), 462 higher in AML and 483 higher in ALL. SNR and t rank genes almost identically ($\rho = 0.999$; Appendix C). The top genes (Appendix, Table IV) agree with known biology: myeloid genes (CST3, CD33, MPO, ZYX, MGST1) are higher in AML, and lymphoid/B-cell genes (TCF3, CD79B, CD19, DNTT, VPREB1, CCND3) are higher in ALL. CST3, ZYX and CD33 are also among the markers highlighted by [1]. Hierarchical clustering of the samples on the top-30 genes (Fig. 3) recovers the classes (adjusted Rand index ARI = 0.89). In Andersson the ARI is only 0.41: the T-ALL samples cluster next to AML, because they lack the B-cell programme that drives most ALL-high genes. With 1,292 samples, 69.1% of the MILE probe sets are significant, clustering again recovers the classes (ARI 0.86), and LEF1 (lymphoid) and MGST1, CFD, APLP2 (myeloid) lead the ranking.

E. Training vs. independent set (batch effect)

The mean Present rate is 29.0% in the training set and 27.3% in the independent set (Mann–Whitney $p = 0.071$). The 75th-percentile intensity, a proxy for array brightness, is lower in the independent set (median 421 vs. 482, $p = 0.016$; Fig. 4a,b). To separate batch from class composition we compared train and test *within* each class with per-gene Kolmogorov–Smirnov tests. Under no shift, 5% of the genes would have $p < 0.05$. We observe 17.7% for ALL (28 genes significant after FDR) and 8.0% for AML (0 genes) (Fig. 4c). So there is a moderate, ALL-specific shift. This is consistent with the more heterogeneous collection of the independent samples, and it is one reason why a single 34-sample test score can differ from CV estimates (**RQ2**).

F. Cross-cohort comparison

On the common genes, the gene-wise t statistics are positively correlated between cohorts (Fig. 5a,b): $\rho = 0.56$ for Golub–MILE, $\rho = 0.44$ for Golub–Andersson and $\rho = 0.67$ for MILE–Andersson. Of the top 100 genes, 33 are shared by Golub and MILE (hypergeometric $p = 3.6 \times 10^{-28}$; 100% with the same direction) and 28 by Golub and Andersson ($p = 1.5 \times 10^{-21}$). 18 genes are in the top 100 of all three cohorts (CST3, CD33, CTSA, POU2AF1, CD79A, PYGL, PCBD1, NPY, ASAH1, MAPKAPK3, CD19, CSTA). The *scales*, however, are incompatible: median expression is 7.31, 0.46 and -0.03 on the three log scales. In a pooled PCA without harmonisation, cohort explains $\eta^2 = 99.8\%$ of PC1 and class only 0.0% (Fig. 5c). After per-cohort gene-wise z-scoring, cohort explains 0.0% and class 61.4% (Fig. 5d). The biology transfers across cohorts but the measurement scale does not, so cross-cohort models need per-cohort standardisation (or ComBat [24]), always estimated without test labels.

G. Numbered findings and implications for modelling

1) *Negative and floor-level intensities* (28.7% < 0) $\Rightarrow$ clip to [20, 16000] and apply $\log_2$ before any scale-sensitive model.

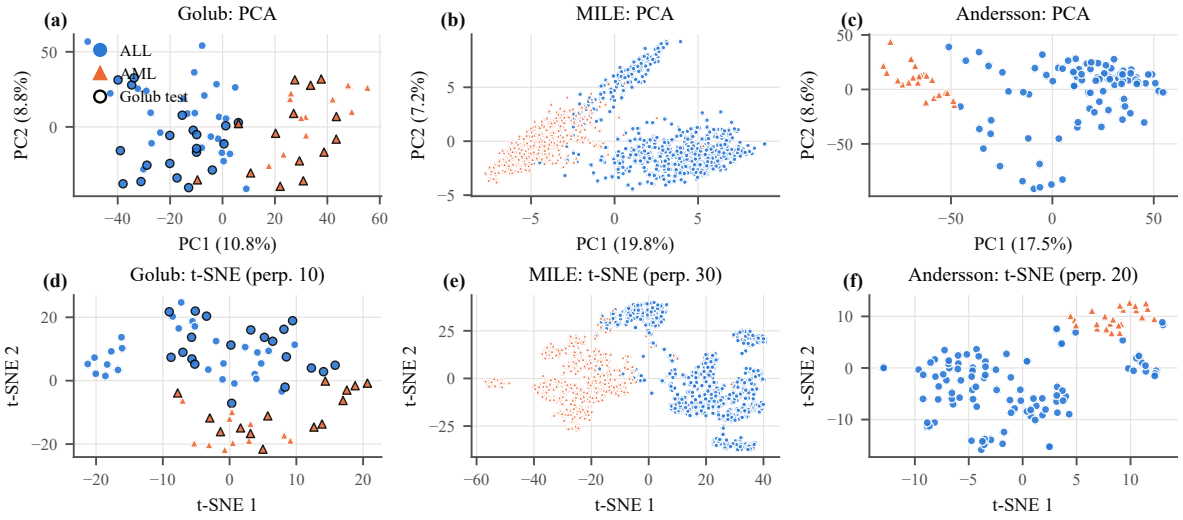


Fig. 2. Unsupervised embeddings of the three cohorts (top: PCA, bottom: t-SNE; colour/marker: class). Golub independent-set samples have black edges. All cohorts show a dominant ALL/AML axis; scree plots are in Appendix C.

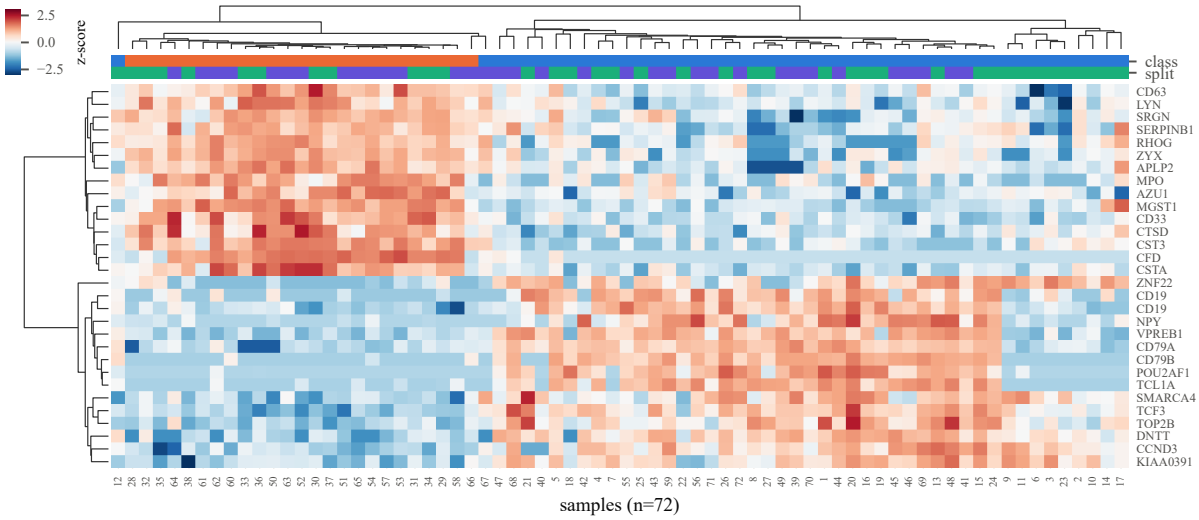


Fig. 3. Golub: z-scored expression of the 15 most AML-high and 15 most ALL-high genes (Welch t), average linkage, $1-$Pearson distance. Column bars: class (blue ALL, orange AML) and split (green train, violet test). Floor-clipped values appear as uniform light-blue blocks.

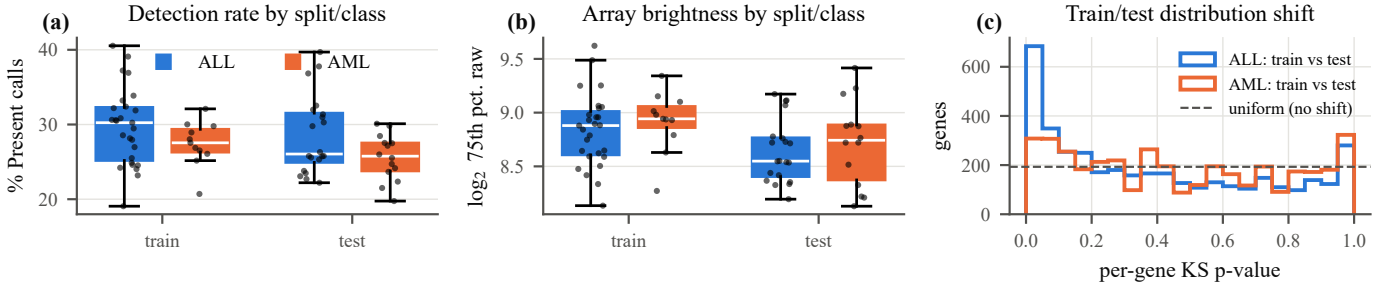


Fig. 4. Golub train vs. independent set. (a) Present-call rate and (b) array brightness by split and class. (c) Per-gene KS p-values comparing train and test within each class; the excess near 0 for ALL indicates a distribution shift.

2) $p \gg n$ (7,129 features, 72 samples) $\Rightarrow$ feature selection (variance or fold-change filter, then univariate k-best) *inside* each training fold, with k tuned in the inner CV. The ablation will cover $k \in \{10, \ldots, 500\}$ and PCA.
3) *Strong low-dimensional signal* (PC1 AUC 0.984) $\Rightarrow$ near-perfect scores are plausible; metric differences be-

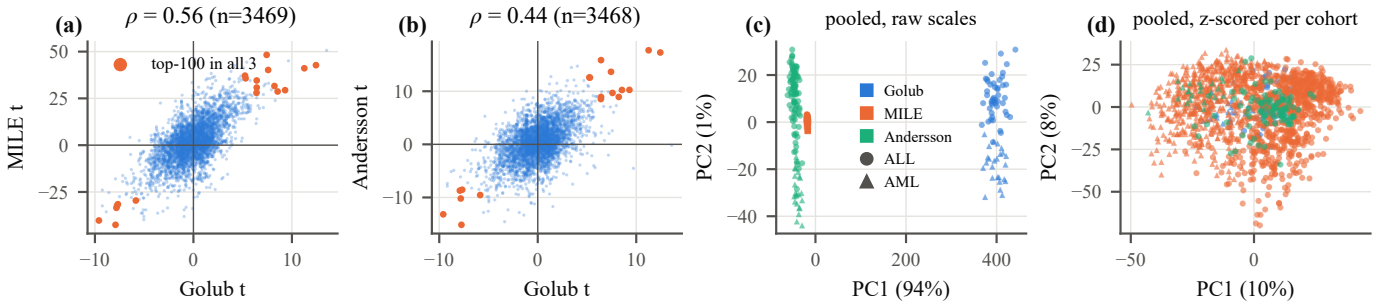


Fig. 5. Cross-cohort compatibility on shared genes. (a,b) Gene-wise Welch t statistics of Golub vs. MILE and vs. Andersson (orange: genes in the top-100 of all three). (c) Pooled PCA without harmonisation separates cohorts, not classes; (d) per-cohort z-scoring brings out the ALL/AML axis (colour: cohort, marker: class).

tween methods will be small, so confidence intervals and paired tests are essential (**RQ1**).

4) *Class imbalance* (65/35, MILE 750/542, Andersson 98/23) ⇒ stratified folds, `class_weight=balanced`, and balanced accuracy, F1 and ROC-AUC alongside accuracy. SMOTE only within training folds, if used at all.
5) *Train/test shift* in Golub ⇒ report the 38/34 result and nested CV separately and explain the difference (**RQ2**).
6) *Missing values* (Andersson) and *scale sensitivity* ⇒ in-fold median/k-NN imputation and in-fold z-scoring for LR, SVM and neural nets (trees are scale-invariant).
7) *Shared biology but different scales* ($\rho > 0$, significant overlap, PC1 dominated by cohort) ⇒ per-cohort standardisation on the 3,469 common genes for cross-cohort transfer (**RQ3**). Lineage heterogeneity (T-ALL) is a likely source of errors and will be analysed.

V. Plan for the Final Report

Methods (no results yet). Three ML methods, chosen to cover different inductive biases: (M1) *elastic-net logistic regression*, a sparse linear model that copes with correlated genes and yields interpretable coefficients [17]; (M2) an *SVM* with linear and RBF kernels, the historical reference for microarrays [10], [11]; and (M3) a *random forest*, which is non-linear, scale-invariant and has built-in importance measures [14], [15]. The module's decision tree is kept as a baseline.

Pipeline and validation. Every data-dependent step of the pipeline (Appendix B, Fig. 6) is a scikit-learn transformer (`src/leukemia/preprocessing.py`) re-fitted inside each fold. Two protocols: (a) the module's 38/34 split, tuned by 5-fold stratified CV on the 38 samples and evaluated *once* on the 34 after model selection; (b) nested CV on all 72 samples, outer 5-fold stratified CV repeated with 5 seeds (42, 7, 123, 2024, 31337), inner 5-fold search, identical outer folds for all methods (paired design). MILE and Andersson use the same protocol; cross-cohort transfer uses the common genes with per-cohort standardisation. **Search spaces:** EN-LR $C \in [10^{-3}, 10^{2}]$ (log), ℓ_1 ratio $\in \{0.1, 0.5, 0.9, 1\}$; SVM $C \in [10^{-3}, 10^{3}]$ (log), $\gamma \in [10^{-5}, 10^{-1}]$; RF #trees $\in \{300, 1000\}$, max-features $\in \{\sqrt{p}, 0.1p, 0.3p\}$, min-leaf $\in \{1, 2, 4\}$; $k \in \{10, \ldots, 500\}$ for all. Grid search (EN-LR, SVM), randomised search (RF) and Optuna TPE (neural nets) will be compared by cost and inner-CV score.

Metrics and statistics. Accuracy, balanced accuracy, precision, recall (sensitivity), specificity, F1, ROC-AUC and confusion matrices, as mean ± SD over 25 outer folds with 95% CIs from the corrected resampled t [3]; Friedman/Nemenyi [4] across methods and McNemar on the fixed split; ROC/PR, loss and k–performance curves.

Bonuses. *B1:* MLP (dropout, weight decay) and autoencoder + classifier, with early stopping and architecture choice in the inner folds and optional pre-training on MILE. *B2:* TabPFN v2 [33] after in-fold selection of ≤ 500 genes, and GenePT-style embeddings [34]. *B3:* submission to an IEEE-sponsored conference open in the final week (e.g. SIU, TIPTEKNO; with the instructor's approval). *B4:* Andersson as the third data set. **Timeline:** weeks 1–2 Golub nested CV (**RQ1**,**RQ2**); 3–4 MILE/Andersson, transfer and ablations (**RQ3**); 5–6 B1/B2 (**RQ4**) and interpretability; 7 paper, video, B3. **Risks:** variance at $n = 72$ (repeated nested CV, CIs); batch/platform effects (per-cohort scaling, ComBat check); neural overfitting (early stopping); TabPFN's feature limit (in-fold selection); LLM memorisation of the famous Golub data (B2).

VI. Reproducibility and Conclusion

`python run_all.py` rebuilds this PDF from raw data (seed 42, pinned `requirements.txt`); every number in the text is a LaTeX macro written by the code. In summary, all three cohorts carry a strong, biologically coherent ALL/AML signal; the obstacles to valid modelling are $p \gg n$, the Golub train/test shift and cross-platform scale differences. The final report will test **RQ1**–**RQ4** with the protocol above.

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Appendix A
Full literature table

TABLE III: Full literature summary table (all 30 included studies, chronological).

Ref.	Year	Data set	Pre-proc./feature sel.	Method	Validation	Best result	Note
[1]	1999	Golub ALL/AML (Affymetrix Hu6800)	Informative-gene selection by class correlation	Class discovery + weighted-vote class predictor	38 train / 34 independent test	n/r	Original dataset; abstract gives no numbers
[10]	2000	Ovarian + two other published sets	n/r	SVM	n/r	n/r	Abstract does not name the other datasets
[5]	2002	MLL/ALL/AML leukemias	n/r	Expression profiling of MLL-rearranged leukemia	n/r	n/r	Common 3-class leukemia benchmark
[11]	2002	Leukemia and colon microarrays	SVM-RFE gene ranking	Linear SVM	n/r	n/r	Canonical wrapper gene selection
[12]	2002	Leukemia + other tumor microarrays	Univariate gene filtering	LDA/DLDA, kNN, trees, aggregation	Repeated train/test splits	n/r	Classic comparison; no abstract retrieved
[13]	2002	SRBCT and leukemia	Shrunken centroid soft-thresholding	Nearest shrunken centroids (PAM)	n/r	n/r	Abstract gives no error counts
[20]	2002	Two published microarray sets	Gene selection inside vs outside CV	Prediction rules with few genes	External 10-fold CV; .632+ bootstrap	n/r	Selection bias; CV error not zero
[16]	2005	11 datasets incl. leukemia	Several gene selection methods	Multicategory SVMs vs kNN/NNs/ensembles	Two cross-validation designs	n/r	MC-SVMs best; GEMS system
[17]	2005	Methodology (incl. microarray example)	Elastic net penalty	Penalized regression	n/r	n/r	Basis for sparse logistic models
[14]	2006	Golub leukemia + 8 other sets	RF backward gene elimination	Random forest vs SVM/kNN/DLDA/SC	.632+ bootstrap	Leukemia .632+ error: SVM 0.014; RF 0.051	Leukemia RF selects 2 genes
[21]	2006	Simulation + microarray	Tuning within CV	Nested CV	Nested vs non-nested CV	n/r	Optimistic bias without nesting
[7]	2007	121 childhood leukemias (GSE7186)	n/r	k-nearest neighbor predictors	Independent data set validation	T-ALL MRD status classified with 100% accuracy	External validation source (pediatric)
[24]	2007	Microarray batches	Empirical Bayes location/scale adjustment	ComBat	n/a	n/r	Standard batch correction
[6]	2008	MILE pre-phase (multi-lab)	Standardized protocols across labs	Inter-laboratory reproducibility study	Cross-lab comparison	n/r	Linked to GSE13159 in GEO
[2]	2010	MILE stage I/II (GSE13159 = stage I)	Whole-genome HG-U133 Plus 2.0 profiling	Gene-expression classifier for leukemia subtypes	Retrospective stage I + prospective stage II cohort	92.2% accuracy over 18 classes (stage I)	External validation source; 11 labs
[22]	2010	Benchmark datasets	Hyperparameter selection	Kernel methods	Model-selection overfitting analysis	n/r	No DOI; JMLR 11:2079-2107
[23]	2010	High-throughput studies (review)	Batch effect diagnostics	Review	n/a	n/r	Motivates cross-cohort batch handling
[8]	2019	CuMiDa (78 curated GEO sets)	Background correction + normalization + QC	Benchmark ML baselines	n/r	n/r	Curated microarray benchmark repository
[25]	2020	12,029 samples from 105 studies	High-dimensional genome-wide input	ML for AML prediction	n/r	n/r	Large-scale blood transcriptome AML prediction
[26]	2021	>16,400 blood transcriptomes (incl. leukaemia)	n/r	Swarm Learning (decentralized ML)	Multi-site with study biases	n/r	Swarm beats single-site classifiers (abstract)
[32]	2021	Tabular benchmarks	n/r	FT-Transformer / ResNet vs GBDT	Benchmark protocol	n/r	No DOI; NeurIPS 34 pp. 18932-18943
[31]	2022	Tabular benchmark suite	n/r	Tree models vs deep learning	Benchmark protocol	n/r	Trees still strong on tabular data
[18]	2022	132 reviewed studies	Feature selection methods	Systematic review	n/a	n/r	Hybrid FS most common (34.9%)
[27]	2022	Golub leukemia (72 x 7129) + 3 others	GNN-based feature ranking aggregation	SVM classifier	10-fold CV	Leukemia Acc 0.971 with 2 genes	From Table 3 of full text
[28]	2022	Golub leukemia (72 x 7129) + DLBCL + prostate	Soft-ensemble wrapper gene selection (kNN)	Stacked deep neural network	K-fold CV	Leukaemia average accuracy 99.6%	Gene selection vs CV order unclear
[29]	2022	Golub leukemia (72 x 7129; UCI)	PSO selected 2323 of 7129 genes	HORD-optimized CNN	Random 70/30 and 80/20 splits	Leukemia accuracy 99.9% (comparison table)	No CV; dataset class description inconsistent
[19]	2023	Microarray and RNA-seq (review)	Feature engineering + preprocessing	Review of ML/DL classifiers	n/a	n/r	Covers MLP/CNN/RNN/GNN/transformers
[30]	2024	Golub ALL/AML + CuMiDa leukemia	RF feature importance (FI >= 0.04)	Super learner (RF meta-learner)	Stratified 10-fold CV with 20% test	AML-ALL accuracy 0.98 (Table 2)	FI ranking appears done before CV
[33]	2025	Tabular datasets up to 10k samples	n/r	TabPFN foundation model	Benchmark comparisons	n/r	Candidate small-n baseline
[34]	2025	Single-cell expression data	GPT-3.5 gene text embeddings	GenePT embedding model	Downstream task benchmarks	n/r	Published GenePT; preprint 10.1101/2023.10.16.562533

Appendix B
Planned pipeline

Appendix C
Additional EDA figures

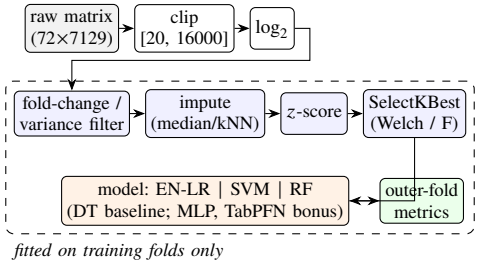


Fig. 6. Planned leakage-free pipeline. Stateless steps (clip, $\log_2$) may precede the split; every step that learns from data (dashed box) is an `sklearn Pipeline` stage re-fitted in each inner and outer training fold.

TABLE IV
TOP 12 GENES BY WELCH p IN EACH COHORT (Δ: MEAN DIFFERENCE AML−ALL ON THE COHORT'S LOG SCALE; $t > 0$: HIGHER IN AML).

	Golub			MILE			Andersson		
#	gene	Δ	t	gene	Δ	t	gene	Δ	t
1	MGST1	+4.10	+13.5	PNKD	+0.25	+56.8	PILRB	-2.95	-16.0
2	CST3	+5.82	+12.4	LEF1-AS1	-0.35	-59.5	INOC1	-1.61	-15.0
3	TCF3	-2.07	-11.8	LEF1	-0.31	-52.5	NPY	-5.62	-15.1
4	ZYX	+3.83	+11.0	CFD	+0.43	+49.8	–	–	–
5	CD33	+2.26	+11.3	APLP2	+0.21	+49.3	NPR1	-3.31	-14.4
6	APLP2	+2.77	+9.6	LOC101928615	+0.22	+48.8	PCBD1	+2.98	+15.9
7	CD79B	-4.45	-9.2	MGST1	+0.38	+53.2	C17orf58	+2.11	+16.7
8	POU2AF1	-3.09	-9.6	MGST1	+0.41	+50.7	SPG20	+3.07	+13.9
9	DNTT	-4.44	-9.4	LEF1	-0.37	-55.1	LOC201229	-3.50	-14.7
10	CCND3	-2.06	-9.2	CTSA	+0.20	+48.2	CTSC	+2.38	+14.8
11	VPREB1	-4.24	-8.6	FUCA2	+0.24	+47.9	PCBD1	+2.99	+14.6
12	RHOG	+3.17	+8.6	LEF1-AS1	-0.25	-47.3	CD33	+3.87	+17.7

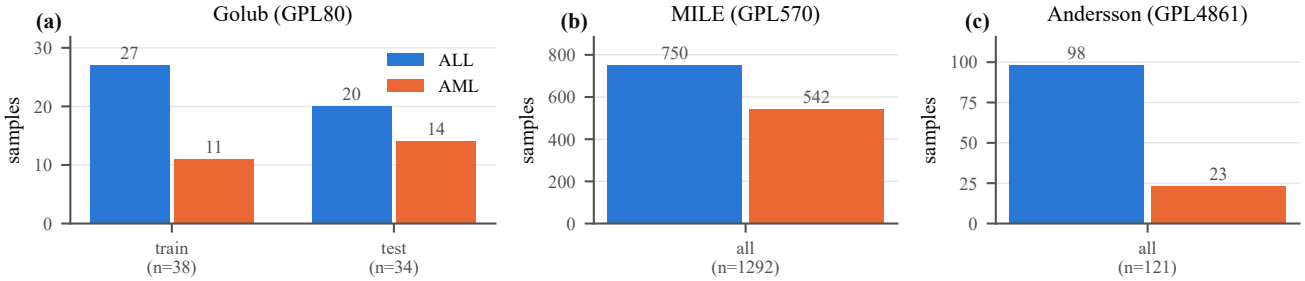


Fig. 7. Class distribution. (a) Golub reference split; (b) MILE after mapping 14 ALL/AML subclasses to two labels; (c) Andersson after removing normal controls. All sets are imbalanced towards ALL.

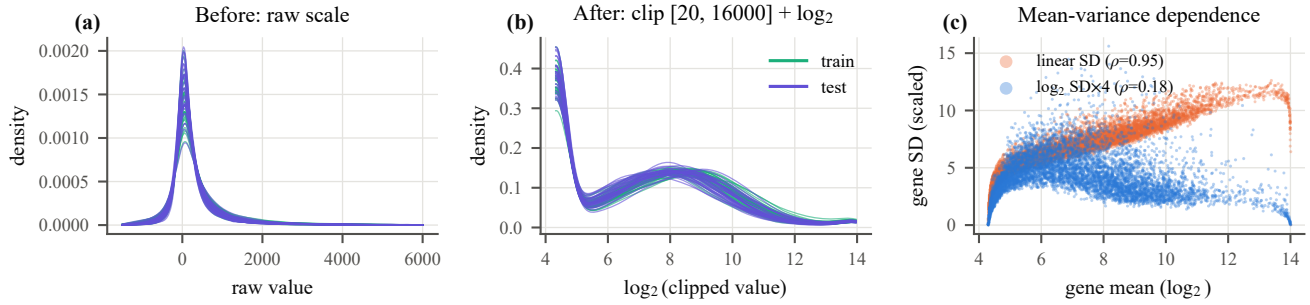


Fig. 8. Golub per-array densities (a) before and (b) after clipping and $\log_2$ (colour: reference split). (c) Gene standard deviation vs. mean: on the linear scale SD grows with the mean ($\rho = 0.95$), after $\log_2$ the dependence largely disappears ($\rho = 0.18$).

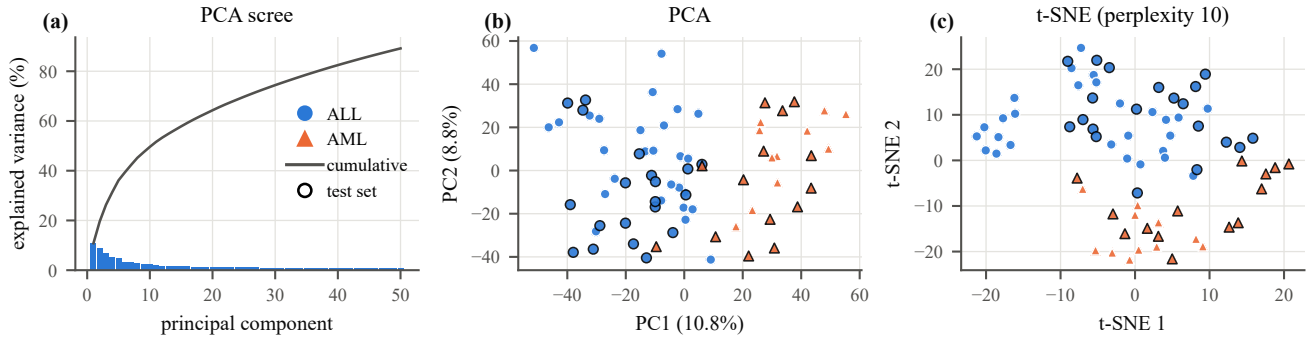


Fig. 9. Golub: PCA scree plot (bars: per component, line: cumulative), PCA and t-SNE. PC1 explains 10.8% and ten PCs 49.6%; 37 PCs are needed for 80%. The variance is spread over many directions, yet PC1 alone separates the classes.

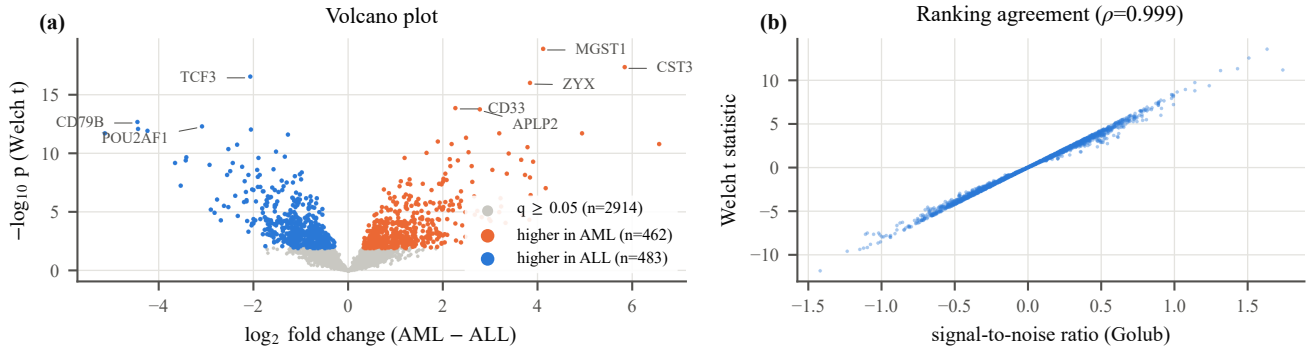


Fig. 10. Golub: (a) volcano plot of Welch tests (colour: BH $q < 0.05$): 462 genes are higher in AML and 483 in ALL, i.e. the differences are not one-sided. (b) Golub's signal-to-noise ratio and Welch t rank genes almost identically ($\rho = 0.999$), so the choice between them matters little for a univariate filter.

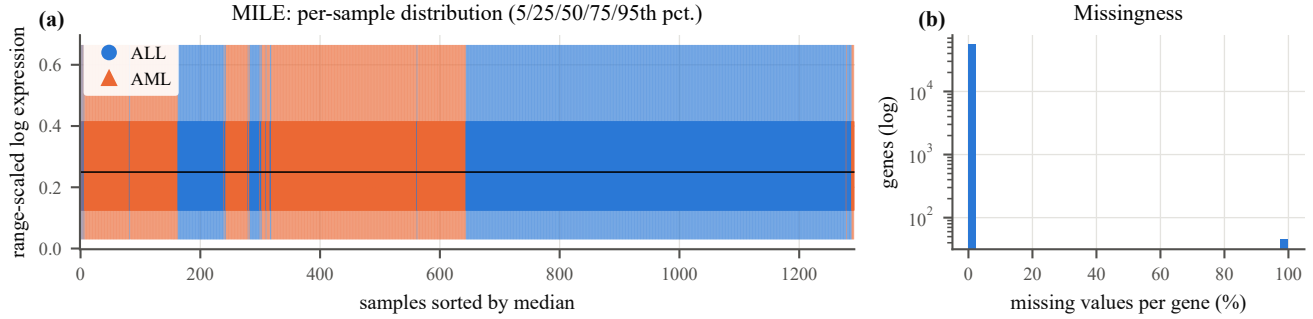


Fig. 11. MILE: (a) per-array quantiles of the range-scaled log values (sorted by median). All arrays have identical quantiles, as expected after quantile normalisation, so there are no array-level outliers. (b) Missing values per probe set: 45 probe sets are empty in every array (removed); all others are complete.

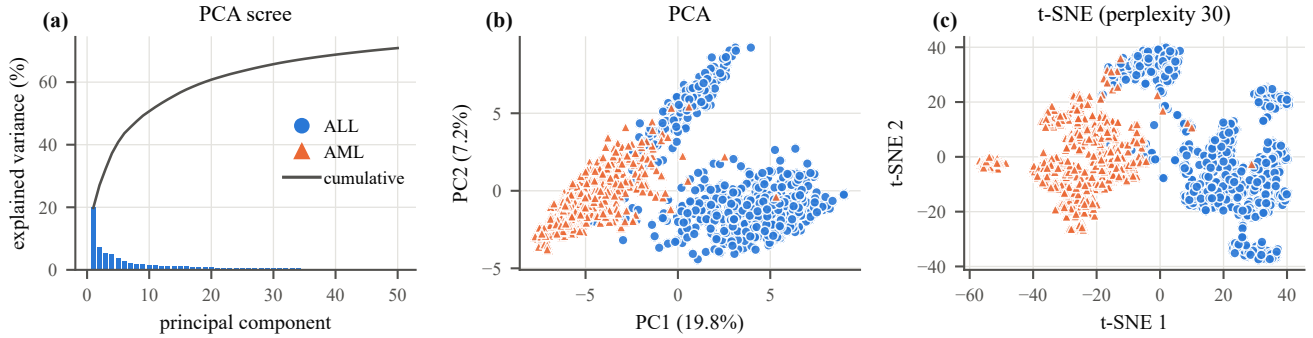


Fig. 12. MILE: PCA scree, PCA and t-SNE (top 5,000 most variable probes). PC1 explains 19.8% and separates ALL from AML (AUC 0.987); in t-SNE the ALL samples form several sub-clusters, in line with the many ALL subtypes in MILE.

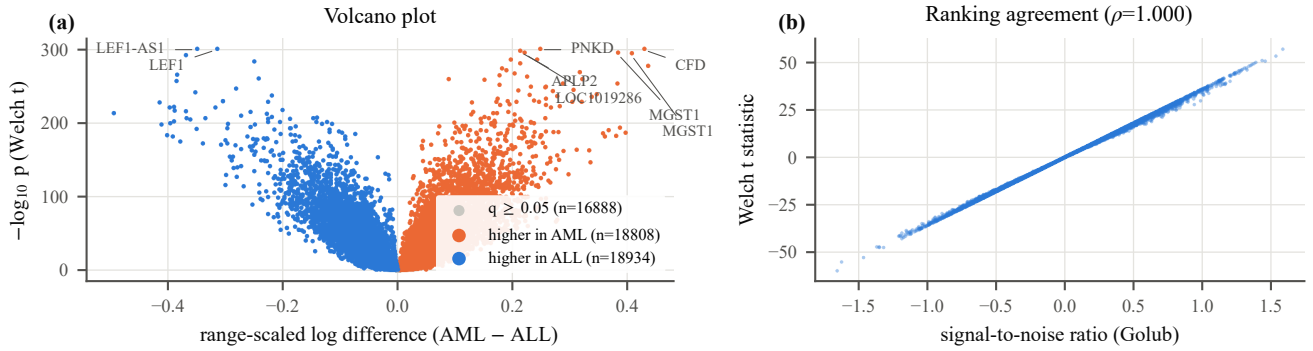


Fig. 13. MILE: (a) volcano plot; the x axis is a difference of range-scaled log values, and $p < 10^{-300}$ is drawn at 300. With $n = 1{,}292$, 69.1% of probe sets reach $q < 0.05$, so effect size rather than p should guide gene selection. (b) SNR–t agreement.

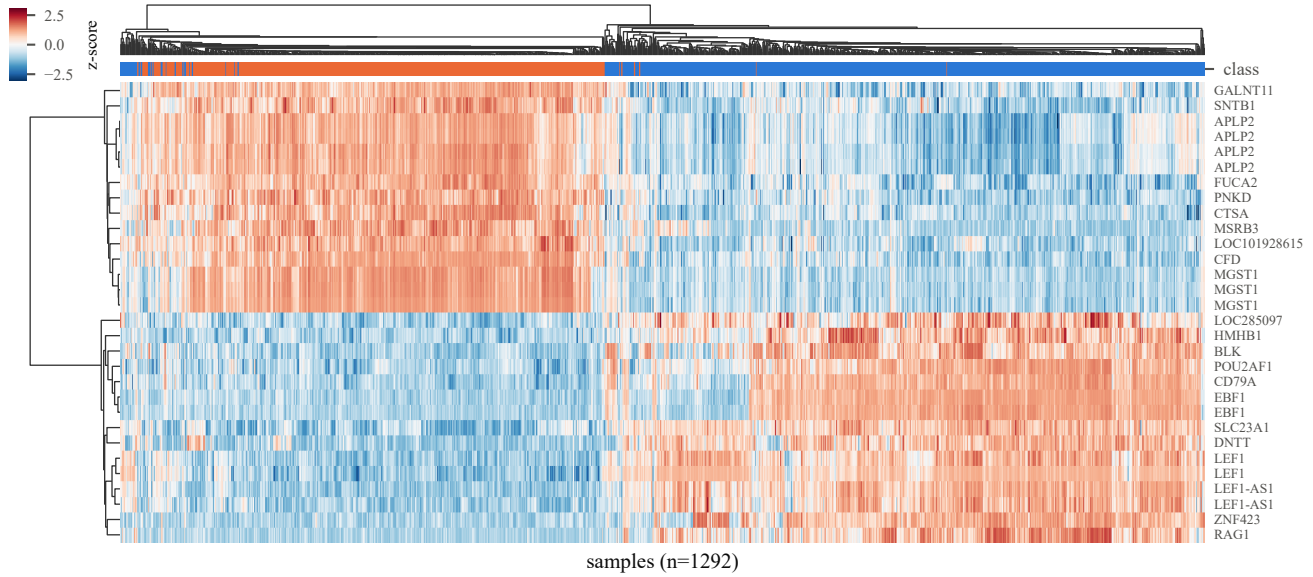


Fig. 14. MILE: clustered heatmap of the top-30 genes. The two top-level clusters match ALL/AML (ARI 0.86); within ALL, a sub-cluster lacks the B-cell genes (EBF1, CD79A, BLK, POU2AF1), consistent with T-ALL.

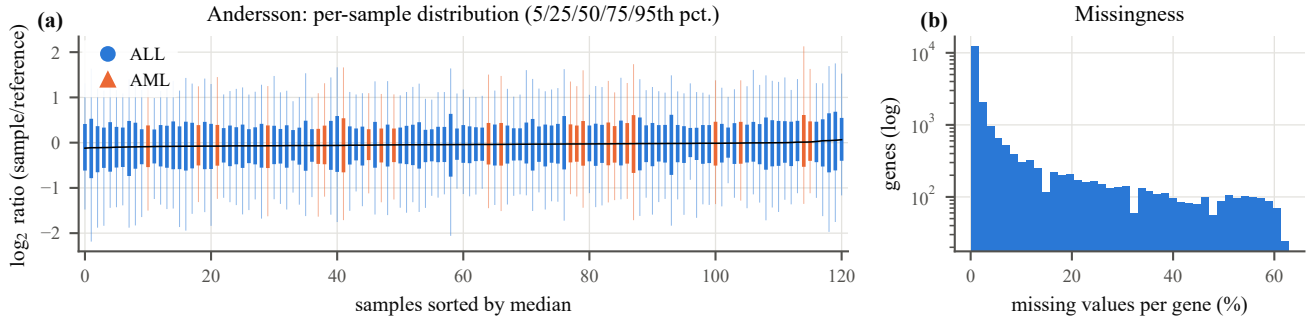


Fig. 15. Andersson: (a) per-array $\log_2$-ratio distributions are centred near 0 with similar spread, as expected after lowess normalisation and centring; (b) missing values per reporter: 12,084 reporters have gaps and 2,841 miss more than 20%, so imputation (in-fold) is required.

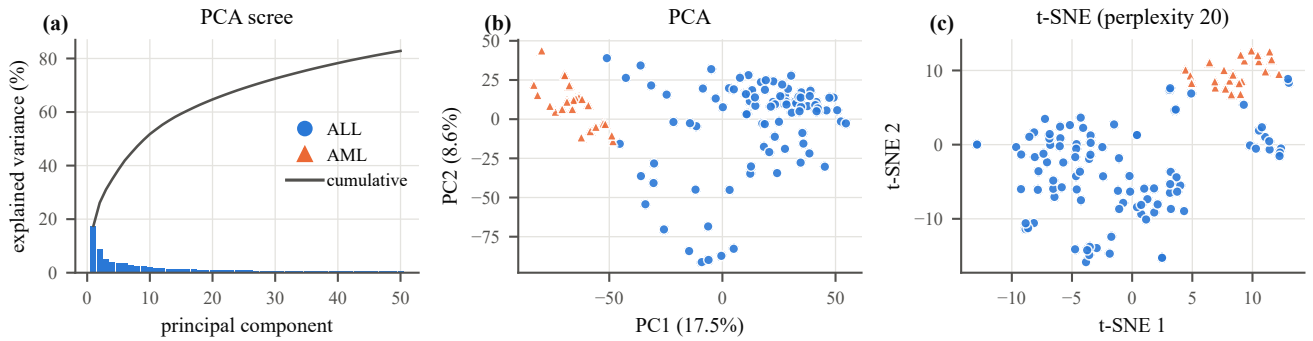


Fig. 16. Andersson: PCA scree, PCA and t-SNE (median-imputed for visualisation only). PC1 explains 17.5% and almost perfectly orders AML vs. ALL (AUC 0.999); t-SNE shows a separate AML group and several ALL sub-groups.

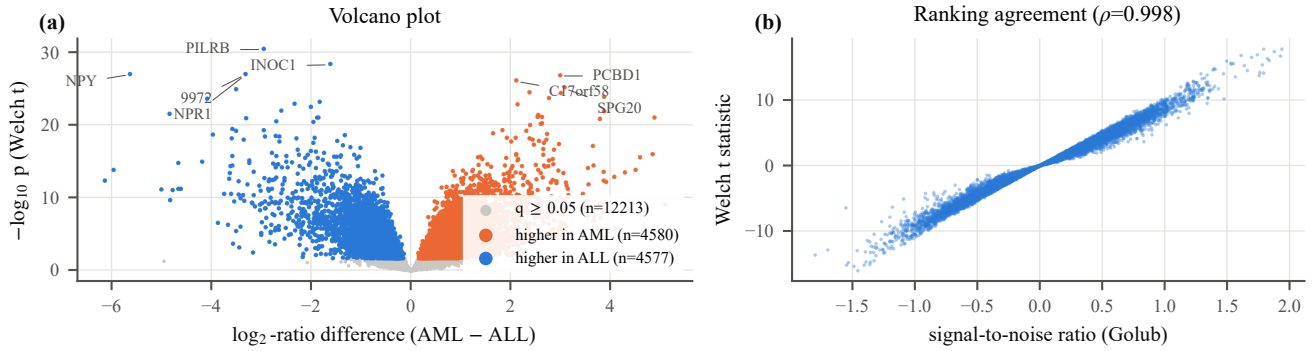


Fig. 17. Andersson: (a) volcano plot (x: difference of $\log_2$ ratios); 42.8% of reporters reach $q < 0.05$, split evenly between AML-high (4,580) and ALL-high (4,577). (b) SNR–t agreement.

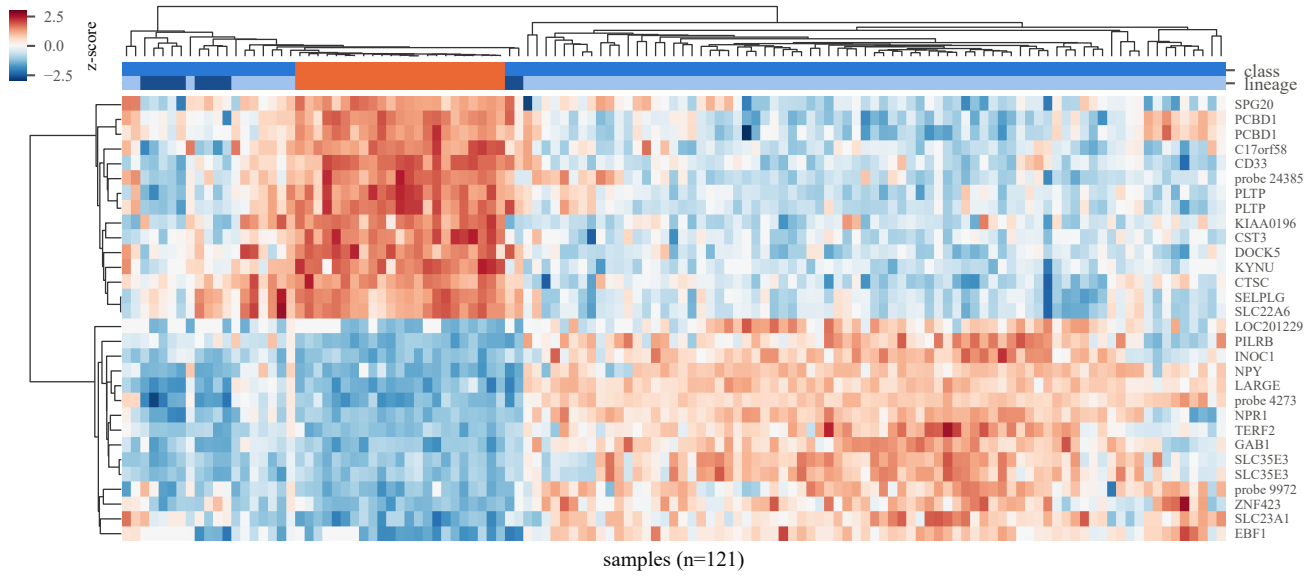


Fig. 18. Andersson: clustered heatmap of the top-30 genes. Lineage bar: light blue B-lineage ALL, dark blue T-cell ALL, orange AML. T-ALL samples cluster next to AML.